DS	**Session**	Tree 2	**Question Information**	Level 1 Challenge 77

Problem Description:

Any sequence *A* of size *n* is called **B-sequence** if:

$A_1 < A_2 < \ldots < A_k > A_{k+1} > A_{k+2} > \ldots > A_n$ where $1 \leq k \leq n$. That is, a sequence which is initially strictly increasing and then strictly decreasing (the decreasing part may or may not be there).

All elements in *A* except the maximum element comes atmost twice (once in increasing part and once in decreasing part) and maximum element comes exactly once.

All elements coming in decreasing part of sequence should have come once in the increasing part of sequence.

You are given a B-sequence **S** and **Q** operations. For each operation, you are given a value **val**. You have to insert **val** in **S** if and only if after insertion, **S** still remains a B-sequence .
After each operation, print the size of **S**. After all the operations, print the sequence **S**.

Hint: Think of using some data structure to support insertion of elements in complexity better than linear.

Input Constraints:

$1 \leq N \leq 10^5$
$1 \leq S_i \leq 10^9$
$1 \leq Q \leq 10^5$
$1 \leq val \leq 10^9$
Given sequence **S** is a **B-sequence**,

Input Format:

First line consists of an integer **N**, denoting size of **S**.
Second line consists of **N** space separated integers, denoting elements of **S**.

Next line consists of an integer **Q**, denoting number of operations.
Each of the following **Q** lines consists of an integer **val**.

Output Format:

After each operation, print the size of **S** in a new line.
After all operations, print the sequence **S**.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
1 2 5 2
4
5
1
3
2
```

EXPECTED OUTPUT

```
4
5
6
6
1 2 3 5 2 1
```

Test Case 2

INPUT (STDIN)

```
5
3 4 5 1 3
3
6
2
3
1
```

EXPECTED OUTPUT

```
6
7
7
1 2 3 4 5 6 3
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=0;i<N;i++)
```

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
8	275	39

Code

b_sequence.c

```c
#include <stdio.h>
#include <stdlib.h>

#define MAXV 1000005

int count_val[MAXV];
int arr[MAXV];

int main() {
    int N;
    if (scanf("%d", &N) != 1) return 0;

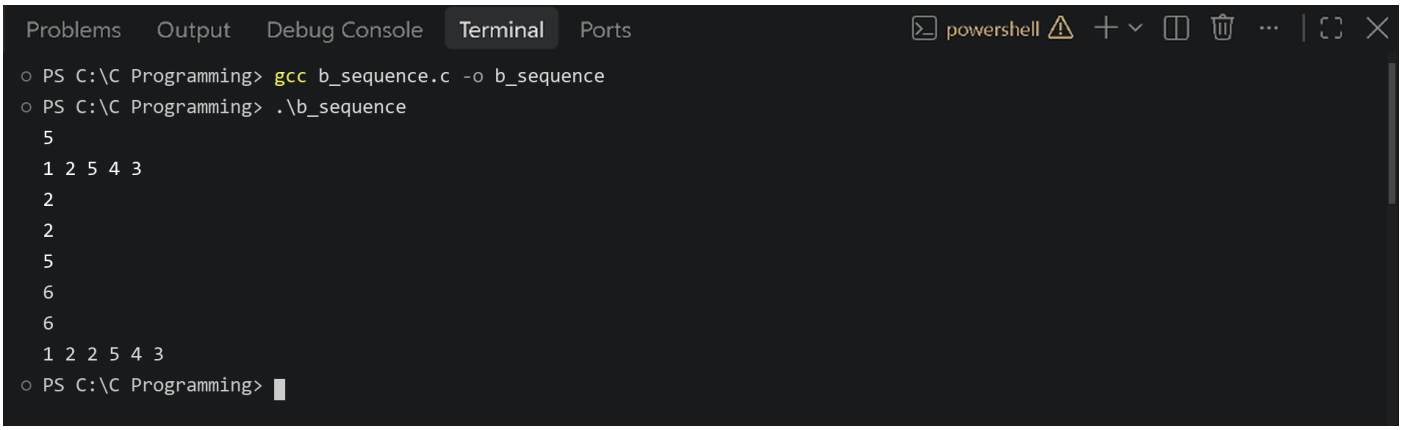
    int max_val = 0;
    int i;
    for(i=0; i<N;i++) {
        scanf("%d", &arr[i]);
        count_val[arr[i]]++;
        if (arr[i] > max_val) max_val = arr[i];
    }

    int current_size = N;
    int Q;
    if (scanf("%d", &Q) != 1) return 0;

    while (Q--) {
        int val;
        scanf("%d", &val);
        if (val > max_val) {
            count_val[val] = 1;
            max_val = val;
            current_size++;
        } else if (val < max_val) {
            if (count_val[val] < 2) {
                count_val[val]++;
                current_size++;
            }
        }
        printf("%d\n", current_size);
    }

    int first = 1;
    for (int v = 1; v <= max_val; v++) {
        if (count_val[v] >= 1) {
            if (!first) printf(" ");
            printf("%d", v);
            first = 0;
        }
    }
    for (int v = max_val - 1; v >= 1; v--) {
        if (count_val[v] == 2) {
            printf(" %d", v);
        }
    }
    printf("\n");
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc b_sequence.c -o b_sequence
PS C:\C Programming> .\b_sequence
5
1 2 5 4 3
2
2
5
6
6
1 2 2 5 4 3
PS C:\C Programming>